IGroupSS-Mamba: Interval Group Spatial–Spectral Mamba for Hyperspectral Image Classification

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***Abstract*— Hyperspectral image (HSI) classification has garnered substantial attention in remote sensing fields. Recent mamba architectures built upon the selective state-space models (S6) have demonstrated enormous potential in long-range sequence modeling. However, the high dimensionality of hyperspectral data and information redundancy pose challenges to the application of S6 in HSI classification, suffering from suboptimal performance and computational efficiency. In light of this, this article investigates a lightweight interval group spatial–spectral mamba framework (IGroupSS-Mamba) for HSI classification, which allows for multidirectional and multiscale global spatial–spectral information extraction in a grouping and hierarchical manner. Technically, an interval group S6 mechanism (IGSM) is developed as the core component, which partitions high-dimensional features into multiple nonoverlapping groups at intervals, and then integrates a unidirectional S6 for each group with a specific scanning direction to achieve nonredundant sequence modeling. Compared with conventional applying multidirectional scanning to all bands, this grouping strategy leverages the complementary strengths of different scanning directions while decreasing computational costs. To adequately capture the spatial–spectral contextual information, an interval group spatial–spectral block (IGSSB) is introduced, in which two IGSM-based spatial and spectral operators are cascaded to characterize the global spatial–spectral relationship along the spatial and spectral dimensions, respectively. IGroupSS-Mamba is constructed as a hierarchical structure stacked by multiple IGSSB blocks, integrating a pixel aggregation-based downsampling strategy for multiscale spatial–spectral semantic learning from shallow to deep stages. Extensive experiments demonstrate that IGroupSS-Mamba significantly outperforms the state-of-the-art methods in classification accuracy and achieves lower model parameters and floating point operations (FLOPs). The code is available at https://github.com/IIP-Team/IGroupSS-Mamba.**

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I. Introduction

HYPERSPECTRAL images (HSIs) are represented by hundreds of continuous spectral bands in the electromagnetic spectrum, spanning from the visible spectrum, near-infrared, mid-infrared to far-infrared. Benefiting from the abundant spectral and spatial information, HSI classification constitutes the fundamental research in remote sensing fields, which allows for the precise pixel-by-pixel identification and detection of materials for ground objects. This technique has demonstrated widespread applications in various remote sensing scenarios, such as urban planning [1], [2], military reconnaissance [3], [4], and precision agriculture [5], [6].

Traditional research on HSI classification predominantly draws upon hand-crafted descriptors and subspace learning methods, such as support vector machine (SVM) [7], linear discriminant analysis (LDA) [8], and manifold learning [9], [10]. To tackle the challenges of spectral variability and spectral confusion, extensive efforts are dedicated to integrating complementary spatial contextual information for more accurate HSI classification, including extended morphological profiles (EMPs) [11], extended multiattribute profiles (EMAPs) [12], and sparse manifold representations [13]. However, these methods generally rely on predefined parameter settings, exhibiting insufficient data fitting and description capabilities.

Benefiting from the powerful nonlinear expressive property and automatic feature extraction capability, deep-learning-based architectures progressively demonstrate predominant status in HSI classification. For instance, Hu et al. [14] first presented a hierarchical 1-D CNN network to extract the high-level spectral features along the spectral dimension of hyperspectral data. Given the significance of spatial correlation in HSIs, Zhong et al. [15] constructed a 3-D CNN spectral–spatial residual network (SSRN), which performs joint spectral–spatial feature extraction with several 3-D convolutional kernels. Although CNN-based models achieve encouraging performance compared to traditional approaches, they generally struggle to capture global contextual characteristics due to inherent local receptive fields. Comparatively, transformer benefits from the powerful long-distance sequence modeling capability based on the attention mechanism, bringing significant paradigms for HSI classification. For example,

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He et al. [16] proposed a bidirectional encoder representation transformer network (HSI-BERT). This architecture captures the correlations between pixels through the self-attention mechanism, which provides the global receptive field and mitigates the restriction of spatial distance. Hong et al. [17] developed a flexible transformer-based SpectralFormer (SF) network, capable of processing both pixelwise and patchwise inputs. This architecture effectively captures spectrally local sequence information from neighboring bands of HSI data, yielding groupwise spectral embeddings (GSEs) for classification. To reduce model complexity and computational burden, Zhao et al. [18] introduced a novel groupwise separable convolutional ViT (GSC-ViT), which incorporates a groupwise separable convolution (GSC) module and a groupwise separable multihead self-attention (GSSA) module. The former partitions HSI data into nonoverlapping groups for individual convolution operations, while the latter involves lightweight groupwise and PWSA mechanisms, significantly reducing the overall parameters quantity. Despite the promising classification results of transformer-based approaches, they typically suffer from quadratic computational complexity $\mathcal{O}(N^2)$ due to the inherent self-attention mechanism, resulting in inefficient modeling and significant memory demands.

Recent Mamba [19] built upon the state-space models (SSMs) establish long-distance dependency through state transitions, which enjoys the advantages of global contextual modeling, linear computational complexity, and selective information processing. For nonsequential vision data, the existing vision mamba (ViM) methods [20] typically extend the Mamba architecture by introducing a multidirectional scanning mechanism. The data scanning is performed horizontally and vertically across spatial dimensions to generate four spatial vision sequences, followed by four independent SSMs for sequence modeling. However, directly applying the existing ViM to HSI classification may encounter challenges in achieving satisfactory performance or computational efficiency due to the following limitations.

1) The projection matrices in Mamba are input-dependent and increase linearly with the number of input channels. The high dimensionality of HSIs inevitably imposes substantial computational burdens.
2) Adjacent spectral bands in original HSIs typically exhibit high similarity, as shown in Fig. 1(a). Traditional multidirectional scanning strategy applied to all the spectral bands may result in information redundancy.
3) Given the abundant spectral information and strong spatial correlation inherent in HSIs, sequence scanning along only the spectral or spatial dimension may lead to the loss of spatial or spectral information, respectively.

In response to the aforementioned challenges, this work proposes a lightweight Interval Group Spatial–Spectral Mamba framework (IGroupSS-Mamba) for HSI classification, enabling multidirectional and multiscale global spatial–spectral information extraction in a grouping and hierarchical manner. Considering the high dimensionality of hyperspectral data and the complementarity of multidirectional scanning of Mamba, an interval group S6 mechanism (IGSM) is introduced. The input high-dimensional features are divided into four nonoverlapping groups at intervals along the spectral dimension, with each group followed by a unidirectional S6 for sequence dependency modeling. After processing, the generated grouping features are concatenated through a channel attention mechanism, addressing the limited interaction inherent in the grouping operation. Following the classic Mamba architecture, IGSM as the fundamental unit is integrated with conventional mapping operations to construct two independent spatial and spectral operators, which are sequentially cascaded and combined with a feedforward network (FFN) to form the interval group spatial–spectral block (IGSSB) for global spatial–spectral feature extraction. Within the spatial and spectral operators, the scanning direction of each group is selected from one of four directions: left-to-right, right-to-left, top-to-bottom, and bottom-to-top, along the spatial and spectral dimensions, respectively. IGroupSS-Mamba is constructed as a hierarchical structure comprising multiple stacked IGSSB blocks, integrated with a pixel aggregation-based downsampling strategy to facilitate the model's multiscale characteristics. The main contributions are summarized as follows.

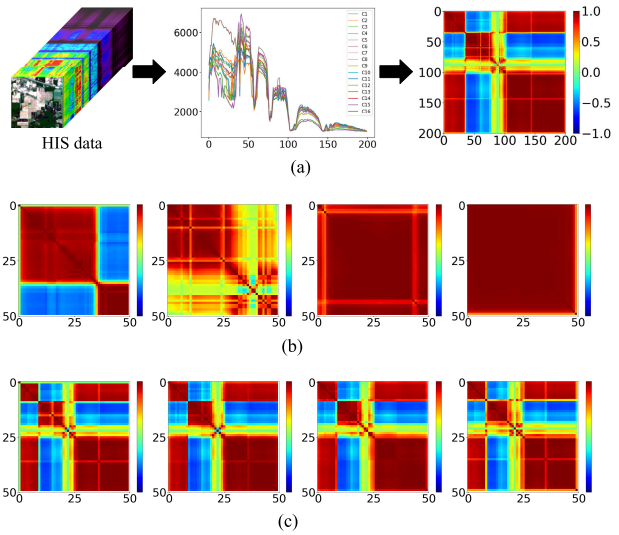


Fig. 1. Spectral correlation matrices of the Indian Pines dataset. (a) Original spectral band. (b) Spectral group generated by adjacent grouping strategy. (c) Spectral group generated by interval grouping strategy.

1) An IGroupSS-Mamba framework based on group modulation and hierarchical mechanism is proposed for HSI classification, which can achieve multidirectional and multiscale global spatial–spectral information extraction with greater computational efficiency.
2) An IGSM is introduced to perform intervalwise feature grouping and parallel unidirectional sequence scanning, which can enjoy the complementarity of different scanning directions while mitigating information redundancy.
3) An IGSSB is designed. By performing IGSM-based grouping sequence modeling along both spatial and spectral dimensions, the spatial regularity and spectral peculiarity of hyperspectral data can be effectively excavated.
4) Extensive experiments are verified on three public hyperspectral datasets: Indian Pines, Pavia University, and

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Houston 2013. The results indicate the effectiveness and superiority of the proposed method.

The remaining sections of this article are organized as follows. Section II reviewed the related work. Section III provides a comprehensive description of the proposed method. Section IV outlines the experimental results and analyses. Conclusions and future work are discussed in Section V.

II. Related Work

A. HSI Classification

The continuous development of deep learning technology has brought substantial paradigms for HSI classification task. Building upon the properties of local receptive fields and parameter sharing, CNN architectures demonstrate progressive innovations. For instance, Hu et al. [14] first proposed a hierarchical 1-D CNN network, which captures the high-level spectral features of hyperspectral data along the spectral dimension. Given the characteristics of abundant spectral channels and strong spatial correlation in HSIs, Lee and Kwon [21] introduced a contextual deep 2-D CNN to jointly leverage both spatial and spectral features for HSI classification. Gao et al. [22] designed an outstanding depthwise feature interaction network (DFINet) that has been proved to be excellent at extracting features for wetland classification in HSIs. In contrast to 2-D CNN extracting features by separating the spatial and spectral dimensions, 3-D convolutional kernels possess the capability for joint spectral–spatial feature extraction. Classically, Hamida et al. [23] developed a 3-D CNN architecture, which captures finer spectral–spatial representations with several stacked 3-D convolutional kernels. The 3-D CNN enables the joint spatial–spectral feature representation from spectral bands, while the 2-D CNN captures abstract-level spatial features. As the mainstream backbone architecture, CNN-based methods exhibit formidable capability in local feature extraction; however, they generally encounter challenges in establishing global dependencies between pixels.

Benefiting from the long-range sequence modeling capability based on attention mechanism, transformer architectures have been significantly investigated for HSI classification task [24], [25], [26], [27], [28], [29]. In particular, He et al. [16] proposed a bidirectional encoder representation transformer network (HSI-BERT), primarily built on a multihead self-attention (MHSA) mechanism in an MHSA layer. Hong et al. [17] constructed a pure transformer-based SF network, which involves a groupwise spectral embedding (GSE) to learn localized spectral representations and a cross-layer adaptive fusion (CAF) to convey features from shallow to deep layers. To comprehensively leverage both spectral and spatial information, Zhong et al. [30] developed a novel spectral–spatial transformer network (SSTN), which consists of spatial attention and spectral association modules to break the long-range constraints of convolution kernels. A factorized architecture search (FAS) model is integrated to determine the layer-level choices and block-level orders. Furthermore, Peng et al. [31] devised a dual-branch spatial–spectral transformer with cross-attention (CASST), in which the spectral branch establishes dependencies among spectral bands and the spatial branch captures mine-grained spatial contexts. The interaction between spatial and spectral information is facilitated within each transformer block via a cross-attention mechanism. To relieve the model complexity and feature overdispersion, Mei et al. [32] introduced a group-aware hierarchical transformer (GAHT), which confines MHSA to the local spatial–spectral relationships by incorporating a grouped pixel embedding (GPE) module. Zhao et al. [18] presented a novel GSC-ViT, which integrates a GSC module to partition HSI data into nonoverlapping groups for distinct convolution operations. A groupwise separable multihead self-attention (GSSA) module comprising groupwise self-attention (GSA) and pointwise self-attention (PWSA) is stacked for local and global spatial feature extraction. While transformer architectures have demonstrated remarkable performance in HSI classification, they suffer from quadratic computational complexity due to inherent self-attention mechanism, presenting substantial challenges in terms of modeling efficiency and memory overhead.

B. SSMs and Mamba

As a foundational model in control theory, the classical state-space model [33] converts higher order derivatives into state variables, allowing dynamic systems to be represented through only first-order derivatives. Building upon this foundation, recent SSMs [34] capitalize on the strengths of continuous state spaces to model complex temporal dynamics, exhibiting significant advantages in long-distance sequence modeling and linear computational complexity. To enhance dynamic modeling capabilities, Mamba [19] introduces a selective SSMs (S6) architecture, integrating time-varying parameters to selectively process relevant information based on input. In addition, a hardware-aware mechanism is integrated to facilitate efficient recursive computation.

Drawing inspiration from the effectiveness of Mamba models in sequence data processing, recent studies have extended its potential to vision scenarios involving 2-D spatial awareness. For example, Zhu et al. [35] proposed a ViM architecture, which performs bidirectional selective scanning in the horizontal direction, transforming the image into ordered sequential data to perceive global information. Liu et al. [20] introduced a VMamba framework that incorporates a cross-selective scan module. The data scanning is performed horizontally and vertically across spatial dimensions to generate four spatial vision sequences, achieving effective global contextual modeling with linear complexity. Furthermore, Huang et al. [36] presented a LocalVMamba model based on windowed selective scan. It introduces a scanning methodology that operates within distinct windows and uses dynamic scanning directions across network layers, effectively capturing local dependencies while maintaining a global perspective. Inspired by these advancements, several researchers have extensively incorporated the Mamba architecture into remote sensing applications, such as classification [37], [38], change detection [39], dense prediction [40], and semantic segmentation [41], [42], significantly highlighting the adaptability and effectiveness of Mamba in handling complex remote sensing signals.

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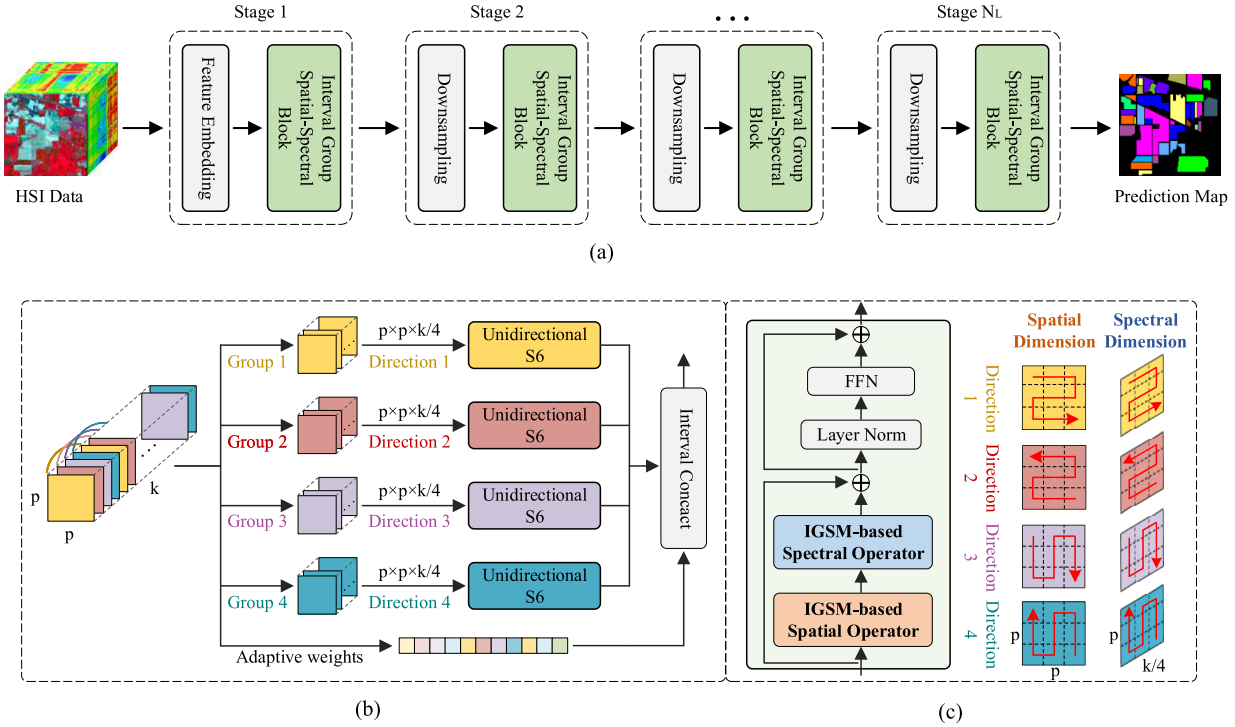


Fig. 2. (a) Overall architecture of the proposed IGroupSS-Mamba framework for HSI classification. (b) Computational procedure of the proposed IGSM. (3) Structural flow of the proposed IGSSB.

III. Proposed Network

This section commences with the preliminaries associated with SSMs. Following this, we investigate the overall architecture of the proposed IGroupSS-Mamba framework for HSI classification. Subsequently, the submodel of IGSM is meticulously introduced, followed by the development of IGSSB.

A. Preliminaries

1) State-Space Models: Drawing from the principles of the linear time-invariant system, SSMs are designed to map a 1-D signal $x(t) \in \mathbb{R}$ into an output sequence $y(t) \in \mathbb{R}$ via an intermediate hidden state $h(t) \in \mathbb{R}^N$. This transformation can be mathematically described through the following linear ordinary differential equation (ODE):

$$\begin{aligned} h'(t) &= \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) &= \mathbf{C}h(t) \end{aligned} \tag{1}$$

where $\mathbf{A} \in \mathbb{R}^{N\times N}$ denotes the state transition parameter, and $\mathbf{B} \in \mathbb{R}^{N\times 1}$, $\mathbf{C} \in \mathbb{R}^{N\times 1}$ represent the projection matrices.

To integrate the continuous-time system depicted in (1) into discrete sequence-based deep models, the continuous parameters $\mathbf{A}$ and $\mathbf{B}$ are subsequently discretized via a zero-order hold (ZOH) technique with a time scale parameter Δ. This process can be expressed as

$$\begin{aligned} \overline{\mathbf{A}} &= \exp(\Delta\mathbf{A}) \\ \overline{\mathbf{B}} &= (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I}) \cdot \Delta\mathbf{B} \\ &\approx (\Delta\mathbf{A})^{-1}(\Delta\mathbf{A})(\Delta\mathbf{B}) \\ &= \Delta\mathbf{B} \end{aligned} \tag{2}$$

where $\overline{\mathbf{A}}$ and $\overline{\mathbf{B}}$ represent the discretized forms of the parameters $\mathbf{A}$ and $\mathbf{B}$, respectively.

After the discretization step, the discretized SSM system can be formulated as follows:

$$\begin{aligned} h_t &= \overline{\mathbf{A}}h_{t-1} + \overline{\mathbf{B}}x_t \\ y_t &= \mathbf{C}h_t. \end{aligned} \tag{3}$$

For efficient implementation, the aforementioned linear recurrence process is achieved through the following convolution operation, which can be expressed as

$$\begin{aligned} \overline{\mathbf{K}} &= \left(\mathbf{C}\overline{\mathbf{B}}, \mathbf{C}\overline{\mathbf{A}\mathbf{B}}, \ldots, \mathbf{C}\overline{\mathbf{A}}^{\mathbf{L}-1}\overline{\mathbf{B}}\right) \\ \mathbf{y} &= \mathbf{x} * \overline{\mathbf{K}} \end{aligned} \tag{4}$$

where L denotes the length of input sequence, and $\overline{\mathbf{K}} \in \mathbb{R}^L$ serves as the structured convolutional kernel.

2) Selective State-Space Models: Traditional SSMs predominantly rely on the simplifying assumption of linear time-invariant, which enjoys the advantage of linear time complexity. However, it generally encounters challenges in capturing contextual relationships due to static parameterizations. To overcome this limitation, Mamba [19] introduces the selective state-space models (S6) to enable the interactions between sequential states. Specifically, S6 implements dynamic adjustments to the projection matrices through sequence-aware parameterization, which can be formulated as

$$\begin{aligned} \mathbf{B} &= \text{Projection}_B(x) \\ \mathbf{C} &= \text{Projection}_C(x) \\ \Delta &= \tau_\Delta(\text{Parameter} + \text{Projection}_\Delta(x)) \end{aligned} \tag{5}$$

i.e., the parameters $\mathbf{B} \in \mathbb{R}^{B\times L\times N}$, $\mathbf{C} \in \mathbb{R}^{B\times L\times N}$, and $\mathbf{\Delta} \in \mathbb{R}^{B\times L\times D}$ are dynamically calculated from the input sequence $x \in \mathbb{R}^{B\times L\times D}$, which facilitates selective perception on each sequence unit.

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B. Interval Group Spatial–Spectral Mamba: Overview

Assuming an original HSI $I \in \mathbb{R}^{H\times W\times V}$, where H and W represent the spatial dimensions, and V denotes the spectral dimension, principal component analysis (PCA) [43] is initially exploited to alleviate the potential Hughes phenomenon [44]. The modified image is represented as $I_{\text{PCA}} \in \mathbb{R}^{H\times W\times L}$, where L refers to the updated spectral dimension after PCA. Given that neighboring pixels can supplement spatial information for the central pixel, the modified image is further partitioned into a series of 3-D patch cubes $\{\text{img} \in \mathbb{R}^{B\times B\times L}\}_{i=1}^{H\times W}$ as input of IGroupSS-Mamba, with labels assigned based on the central pixel of each patch.

The architecture of the proposed IGroupSS-Mamba for HSI classification is illustrated in Fig. 2(a). Adhering to the hierarchical architecture in VMamba [20], IGroupSS-Mamba is structured into multiple stages, with each stage comprising a downsampling operation (replaced by pixel embedding (PE) in the first stage) and an IGSSB. Initially, the cropped patch undergoes PE to preserve fine-grained information. Subsequently, the generated shallow features are processed through multiple stacked IGSSB blocks, with downsampling operations integrated to facilitate multiscale semantic representation. Ultimately, the extracted spatial–spectral features are transmitted to the prediction module for final classification.

1) Pixel Embedding: This process is implemented with one 3-D convolutional block, followed by an embedding operation. Taking the cropped cube img as input, the procedure can be formulated as

$$f = \Phi_{\text{embed}}(\Phi_{\text{3DConv}}(\text{img})) \tag{6}$$

where f denotes the generated shallow spatial–spectral features. The 3-D convolution block consists of a 3-D convolution layer, a batch normalization layer, and a ReLU activation function. The embedding operation involves a linear layer for dimension transformation.

2) Downsampling: The adjacent pixels within a homogeneous region in HSIs generally exhibit similar textural information. Based on this, the downsampling operation is achieved by performing pixel aggregation along the spatial dimension. Given $f_i \in \mathbb{R}^{P_i\times P_i\times K_i}$ as the input features in i $\{i = 2, 3, \ldots, N_L\}$ stage, the downsampling process can be formulated as

$$f_i = \Phi_{\text{embed}}(\text{Aggr}(f_i, \varepsilon = \{m \times m, s\})) \tag{7}$$

where ε signifies the downsampling scale, defined by the aggregation kernel size $m \times m$ and the mobility stride s. The aggregation operation Aggr$(\cdot)$ is performed through pixel averaging. Similar to (6), a linear embedding layer is followed to achieve feature mapping with the dimension of d_i. Consequently, the corresponding output of the ith stage yields the resolution of $p_i \times p_i \times k_i$, where $p_i = (((P_i - m)/s) + 1)$ and $k_i = d_i$.

The downsampling operation adaptively forms multiscale features within the hierarchical architecture, enabling IGroupSS-Mamba to effectively model long-range dependencies with richer contextual semantics and fewer computational costs.

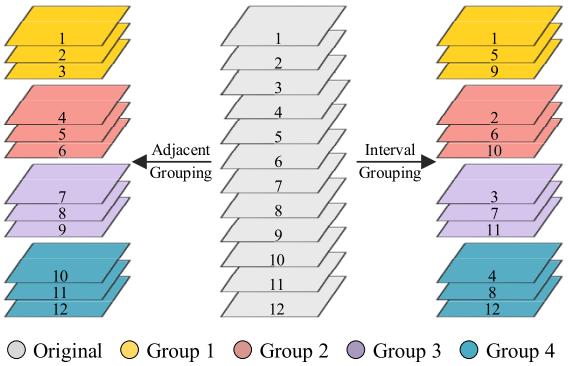


Fig. 3. Example of adjacent grouping strategy (left) and interval grouping strategy (right).

3) Interval Group Spatial–Spectral Block: As the core component of IGroupSS-Mamba, the IGSSB is responsible for capturing global spatial–spectral contextual information based on the IGSM. After PE or downsampling in each stage, the mapped features are fed into the IGSSB block. This procedure can be delineated as follows:

$$f_i^{\text{out}} = \text{IGSSB}(f_i). \tag{8}$$

The output feature map f_i^{out} retains the same resolution as the input. More detailed descriptions of IGSM and IGSSB are provided in Sections III-C and III-D.

After modeling with N_L stages, the extracted spatial–spectral features $f_{N_L}^{\text{out}}$ undergo an average pooling operation, and then pass through the Classifier$(\cdot)$ comprising one perceptron layer to yield the ultimate classification results

$$\text{pred} = \text{Classifier}\left(\Phi_{\text{avg}}\left(f_{N_L}^{\text{out}}\right)\right). \tag{9}$$

C. Interval Group S6 Mechanism

HSIs are represented by hundreds of continuous spectral bands, typically exhibiting high dimensionality and significant similarity between adjacent bands. Applying traditional multidirectional scanning strategies to all spectral bands will inevitably result in substantial computational demand and information redundancy. To this end, this article proposes an IGSM, as illustrated in Fig. 2(b). The key idea is to enable parallel sequence modeling by constructing multiple nonoverlapping feature groups, each processed with a specific unidirectional scanning direction. The IGSM primarily consists of three stages: feature interval grouping, unidirectional S6 for sequence modeling, and feature interval concatenation.

1) Interval Grouping: As illustrated in Fig. 3, unlike traditional adjacent grouping strategy that relies on continuous local cubes, the interval grouping strategy takes into account the similarity between adjacent spectral bands. Taking $F = \{[r_1, r_2, \ldots, r_k]\} | r_i \in \mathbb{R}^{p\times p\times 1}$ as the input high-dimensional feature, it is partitioned into four nonoverlapping groups along the spectral dimension in an interval manner, which can be formulated as

$$G_i = \left\{\left[r_i, r_{i+4}, \ldots, r_{k+i-4}\right]\right\}, \quad i = 1, 2, 3, 4. \tag{10}$$

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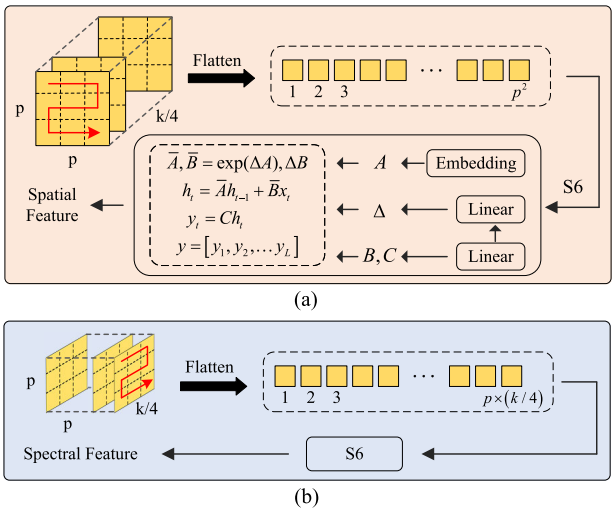


Fig. 4. Specific computation procedure for unidirectional S6, taking the scanning direction of left-to-right as an example. (a) Spatial dimension. (b) Spectral dimension.

The dimensionality of spectral channels for each group is reduced to one-quarter of the input dimensionality, which enjoys the resolution of $G_i \in \mathbb{R}^{p\times p\times(k/4)}$. Interval grouping facilitates the construction of nonredundant global information within each group, while reducing data dimensionality to enhance computational efficiency.

2) Unidirectional S6: To leverage the complementarity of multidirectional scanning, each feature group is followed by a unidirectional S6 for sequence modeling, with the scanning direction selected from one of four routes: left-to-right, right-to-left, top-to-bottom, and bottom-to-top, i.e., υ_{LR}, υ_{RL}, υ_{TB} and υ_{BT}. Considering the intricate spectral correlations and spatial patterns in HSI data, we construct these four directions from both spatial and spectral dimensions, which are, respectively, applied to the spatial and spectral operators within IGSSB, as depicted in Fig. 2(c). As a result, taking the grouping features G_1, G_2, G_3, and G_4 as input, the parallel modeling procedure for each group can be formulated as

$$\begin{aligned} G_1^{\mathrm{out}} &= S6(G_1, d_1 = \{\upsilon_{LR}, \omega_{\mathrm{dim}}\}) \\ G_2^{\mathrm{out}} &= S6(G_2, d_2 = \{\upsilon_{RL}, \omega_{\mathrm{dim}}\}) \\ G_3^{\mathrm{out}} &= S6(G_3, d_3 = \{\upsilon_{TB}, \omega_{\mathrm{dim}}\}) \\ G_4^{\mathrm{out}} &= S6(G_4, d_4 = \{\upsilon_{BT}, \omega_{\mathrm{dim}}\}) \end{aligned} \tag{11}$$

where ω_{dim} refers to either the spatial or spectral dimension. Taking the first group as an example, the specific computation procedure for unidirectional S6 is illustrated in Fig. 4.

Spatial Dimension: The grouping feature $G_1 \in \mathbb{R}^{p\times p\times(k/4)}$ first undergoes pixelwise flattening from left to right along the spatial dimension, generating the flattened 1-D sequence $G_1^{\mathrm{spa}} = \{[x_0, x_1, \ldots, x_{p^2}]\}|x_j \in \mathbb{R}^{1\times(k/4)}$. Subsequently, this sequence is transmitted into the S6 model for state-space evolution as

$$\begin{aligned} h_j &= \overline{\mathbf{A}}_{\mathrm{spa}} h_{j-1} + \overline{\mathbf{B}}_{\mathrm{spa}} x_j \\ y_j &= \mathbf{C}_{\mathrm{spa}} h_j \end{aligned} \tag{12}$$

where $\overline{\mathbf{A}}_{\mathrm{spa}}$, $\overline{\mathbf{B}}_{\mathrm{spa}}$, and $\mathbf{C}_{\mathrm{spa}}$ are the training parameters of the S6 model, and N denotes the hidden state dimension. After efficient computation, the output sequence $G_1^{\mathrm{out}} = \{[y_0, y_1, \ldots, y_{p^2}]\}|y_j \in \mathbb{R}^{1\times(k/4)}$ is obtained for subsequent processing. Notably, the same procedure is executed for the remaining three parallel groups, with different scanning directions and independent training parameters.

Spectral Dimension: As illustrated in Fig. 4(b), the grouping feature $G_1 \in \mathbb{R}^{p\times p\times(k/4)}$ performs data flattening from left to right along the spectral dimension, yielding the flattened spectral sequence $G_1^{\mathrm{spe}} = \{[x_0, x_1, \ldots, x_{p\times(k/4)}]\}|x_j \in \mathbb{R}^{1\times p}$. Similarly, this sequence is subsequently transmitted into the S6 model as

$$\begin{aligned} h_j &= \overline{\mathbf{A}}_{\mathrm{spe}} h_{j-1} + \overline{\mathbf{B}}_{\mathrm{spe}} x_j \\ y_j &= \mathbf{C}_{\mathrm{spe}} h_j \end{aligned} \tag{13}$$

where $\overline{\mathbf{A}}_{\mathrm{spe}}$, $\overline{\mathbf{B}}_{\mathrm{spe}}$, and $\mathbf{C}_{\mathrm{spe}}$ represent the training parameters. The ultimate transformed output can be expressed as $G_1^{\mathrm{out}} = \{[y_0, y_1, \ldots, y_{p\times(k/4)}]\}|y_j \in \mathbb{R}^{1\times p}$.

3) Feature Interval Concatenation: After parallel group processing, the generated grouping outputs are subsequently subjected to interval concatenation. To encourage intergroup information interaction, dynamic weights are modulated in response to the spectral channels as

$$W = \sigma\left(\Phi_{\mathrm{atten}}\left(\Phi_{\mathrm{AvgPool}}(F)\right)\right) \tag{14}$$

where Φ_{AvgPool} represents the average pooling operation to calculate channel statistics. Φ_{atten} consists of two linear layers designed to learn the attention weights for each channel, in conjunction with the nonlinearity function σ. Subsequently, the dynamic weights are leveraged to recalibrate the output grouping features, which can be formulated as

$$F^{\mathrm{out}} = W \cdot \mathrm{InterCat}\left(G_1^{\mathrm{out}}, G_2^{\mathrm{out}}, G_3^{\mathrm{out}}, G_4^{\mathrm{out}}\right). \tag{15}$$

Here, InterCat$(\cdot)$ denotes the intervalwise concatenation operation, which enables each spectral channel to maintain their positional order consistent with the original input.

D. Interval Group Spatial–Spectral Block

To effectively exploit the inherent spectral peculiarity and spatial regularity in HSI data, IGSSB integrates both IGSM-based spatial and spectral operators in a cascaded manner, enabling the extraction of global spatial–spectral semantic representations. The detailed structure of IGSSB is illustrated in Fig. 5.

Letting f_i represent the input features of the ith stage, the forward process for IGSSB can be mathematically described as

$$\begin{aligned} f'_i &= f_i + \mathrm{Spe}(\mathrm{Spa}(f_i)) \\ f_i^{\mathrm{out}} &= f'_i + \mathrm{FFN}\left(f'_i\right) \end{aligned} \tag{16}$$

where Spa$(\cdot)$ and Spe$(\cdot)$ denote the IGSS-based spatial and spectral operators, respectively. The intermediate representation f'_i is transmitted to the FFN$(\cdot)$ comprising two linear layers and an activation function, which enhances the representation capability of model through nonlinear transformations.

Following the classic Mamba architecture [20], the spatial operator Spa$(\cdot)$ and spectral operator Spe$(\cdot)$ perform with

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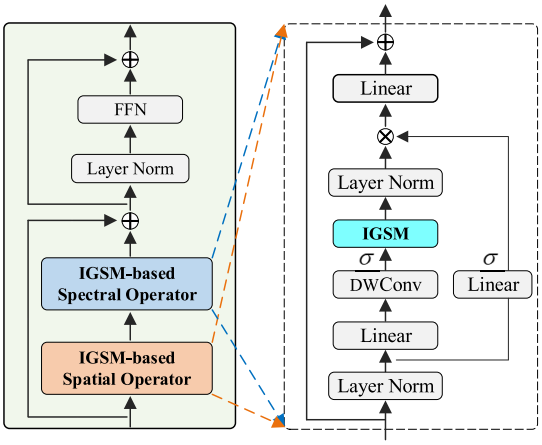


Fig. 5. Detailed structure of IGSSB.

consistent forward procedure, as illustrated on the right side of Fig. 5.

Specifically, the operator commences with a normalization layer to enhance model training stability. Following this, two parallel linear embedding layers are stacked: one branch is followed by an activation function for gating signal generation, while the other branch undergoes a depthwise convolution (DWConv) operation. The procedure can be formulated as

$$z = \sigma(\Phi_{\text{linear}}(\Phi_{\text{norm}}(f_i))) \tag{17}$$

$$F = \sigma(\Phi_{\text{DWConv}}(\Phi_{\text{linear}}(\Phi_{\text{norm}}(f_i)))) \tag{18}$$

where σ denotes the Silu [45] activation operation. After this, the generated representation passes through the pivotal IGSM layer, executing the group-based sequence scanning as described in Section III-C. The output subsequently undergoes layer normalization and a gating operation. Finally, the features are transmitted to the ultimate linear layer, followed by a residual connection:

$$R = \Phi_{\text{linear}}(\Phi_{\text{norm}}(\text{IGSM}(F)) \otimes z) + f_i. \tag{19}$$

The IGSM$(\cdot)$ layer serves as the core computational unit, with the scanning directions for the spatial and spectral operators illustrated in Fig. 2(c). Both perform left-to-right, right-to-left, top-to-bottom, and bottom-to-top for the groups, but along the spatial and spectral dimensions, respectively. The output of IGSSB maintains a consistent resolution as input.

The procedural steps of the proposed IGroupSS-Mamba are summarized in Algorithm 1.

Algorithm 1 Pseudo Procedure of the Proposed IGroupSS-Mamba

Input: Input HSI data $I \in \mathbb{R}^{H\times W\times V}$, patch size B, stage depth N_L.

Output: Predicted labels of the testing samples.

Apply PCA for dimension reduction $I_{PCA} \in \mathbb{R}^{H\times W\times L}$, with $L = 30$.
Crop patch $img \in \mathbb{R}^{B\times B\times L}$, with labels determined by the central pixel.
for depth $i = 1$ to N_L **do**
  if $i = 1$ **then**
    Perform features embedding on img based on (6), generating shallow spatial–spectral features.
  else
    Perform pixel aggregation on f_i based on (7) for downsampling.
  end if
  IGSSB block:
  for {Spatial Operator, Spectral Operator} **do**
    Apply linear embedding on normalized f_i based on (17), generating gating signal z.
    Apply linear embedding and depthwise convolution on normalized f_i based on (18), generating feature F.
    IGSM mechanism:
    Perform interval feature grouping on F based on (10), generating nonoverlapping feature groups G_1, G_2, G_3, and G_4.
    Perform parallel S6 sequence modeling for each group based on (11).
    Perform interval feature concatenation based on (15), generating output F^{out}.
    Apply layer normalization and gating operation based on (19).
  end for
  Apply FFN operation based on (16), generating the output features f_i^{out}.
end for
Classifier:
Perform ultimate classification on spectral–spatial feature $f_{N_L}^{out}$ based on (9), generating results $pred$.

IV. Performance Evaluation

A. Datasets' Description

To validate the classification capabilities of the proposed IGroupSS-Mamba, we conduct comprehensive experimental comparisons on three publicly HSI databases: Indian Pines, Pavia University, and Houston 2013. The division details for training and testing sets are provided in Table I.

1) Pavia University: The dataset was acquired by the reflective optics system imaging spectrometer (ROSIS) over Pavia, Northern Italy. The imaging wavelength of the spectrometer ranges from 0.43 to 0.86 μm. The dataset encompasses 103 spectral bands (after removing 12 noisy bands) and 610 × 340 pixels, with a spatial resolution of 1.3 m per pixel. There are a total of 42 776 ground object pixels, covering nine distinct categories including Asphalt, Gravel, etc.

2) Indian Pines: The dataset was captured by the airborne/visible infrared imaging spectrometer (AVIRIS) imaging an Indian pine tree over Northwestern Indiana in 1992. The imaging wavelength of the spectrometer ranges from 0.4 to 2.5 μm. The dataset consists of 200 spectral bands (after removing the water absorption channels) and 145 × 145 pixels, with a spatial resolution of 20 m per pixel. There are 10 249 ground sample points, representing 16 distinct types including Alfalfa, Corn-notill, etc.

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TABLE I
CATEGORY NAME, SAMPLE NUMBERS OF TRAIN SET AND TEST SET OF EACH CLASS ON THE INDIAN PINES, PAVIA UNIVERSITY, AND HOUSTON 2013 DATASETS

	Indian Pines			Pavia University			Houston 2013		
No.	Category	Train	Test	Category	Train	Test	Category	Train	Test
1	Alfalfa	5	41	Asphalt	332	6299	Healthy grass	125	1126
2	Corn-notill	143	1285	Meadows	932	17717	Stressed grass	125	1129
3	Corn-mintill	83	747	Gravel	105	1994	Synthetic grass	70	627
4	Corn	24	213	Trees	153	2911	Trees	124	1120
5	Grass-pasture	48	435	Painted metal sheets	67	1278	Soil	124	1118
6	Grass-trees	73	657	Bare soil	251	4778	Water	33	292
7	Grass-pasture-mowed	3	25	Bitumen	67	1263	Residential	127	1141
8	Hay-windrowed	48	430	Self-blocking bricks	184	3498	Commercial	124	1120
9	Oats	2	18	Shadows	47	900	Road	125	1127
10	Soybean-notill	97	875				Highway	123	1104
11	Soybean-mintill	245	2210				Railway	123	1112
12	Soybean-clean	59	534				Parking Lot 1	123	1110
13	Wheat	20	185				Parking Lot 2	47	422
14	Woods	126	1139				Tennis Court	43	385
15	Buildings	39	347				Running Track	66	594
16	Stone	9	84						
	Total	1024	9225	Total	2138	40638	Total	1502	13527

3) Houston 2013: The dataset was collected by the ITRES CASI-1500 sensor over the University of Houston campus and its surrounding areas, provided by the 2013 GRSS Data Fusion Contest. The imaging wavelength of the spectrometer ranges from 0.38 to 1.05 μm. The dataset comprises 144 spectral bands and 340 $\times$ 1905 pixels, with a spatial resolution of 2.5 m per pixel. The total 15 029 sample pixels are categorized into 15 challenging types, such as Healthy grass, Trees, etc.

B. Experimental Settings

1) Evaluation Metrics: Following the existing HSI classification approaches, overall classification accuracy (OA), average classification accuracy (AA), and kappa coefficient (Kappa) are exploited as the evaluation metrics. All the experiments are performed under identical experimental conditions, with the reported results averaged over five consecutive trials.

2) Implementation Details: All the experiments are conducted on the PyTorch platform, using a single RTX 3070Ti GPU. For parameter optimization, the Adam optimizer is used with a learning rate set at 0.001. The training process is configured with 100 epochs and a batch size of 64. The PCA dimension for reduction is set to 30. Consistent with the default hyperparameters outlined in VMamba [20], the state dimension and expansion ratio in S6 mechanism are fixed at 16 and 1, respectively.

3) Competitive Approaches: To demonstrate the effectiveness of the proposed IGroupSS-Mamba, three kinds of representative HSI classification architectures are selected for comprehensive comparison, including conventional methods (SVM [7]), CNN-based methods (2D-CNN, 3D-CNN [23], and transformer-based methods (HSI-BERT [16], SF [17], SSTN [30], CASST [31], GSC-ViT [18], SS-MTr [46]).

SVM: The algorithm is implemented using the radial basis function (RBF) kernel on the libsvm toolbox platform. For RBF kernel, the two hyperparameters, σ and λ, are optimally determined through fivefold cross-validation on the training set, with σ ranging from $[2^{-3}, 2^{-2}, \ldots, 2^4]$ and λ from $[10^{-2}, 2^{-1}, \ldots, 10^4]$.

2-D CNN: This architecture consists of two 2-D convolutional blocks and one fully connected (FC) linear layer. Each convolutional block includes a 2-D convolutional layer, a ReLU activation function, and a max-pooling layer. The feature dimensions for the two convolutional layers are set to 8 and 32, respectively, with a kernel size of 3 $\times$ 3.

3-D CNN: This model stacks two 3-D convolution blocks followed by an FC layer. The feature dimensions of the two 3-D convolutional layers are set to 8 and 16, respectively, with a kernel size of 3 $\times$ 3 $\times$ 3.

HSI-BERT: This model consists of a PE block for vector transformation and two stacked bidirectional transformer-based encoders. The extracted features are fed into an FC linear layer for label prediction.

SF: This architecture is designed to capture spectral sequence attributes, consisting of five transformer encoders and each incorporating a CAF module. The feature embedding size for each encoder is fixed at 64.

SSTN: The model comprises two stacked spatial attention modules and two spectral association modules, using a factorization architecture search (FAS) to determine the hierarchical operations and block-level ordering.

CASST: This model is designed as a dual-branch spatial–spectral transformer framework, comprising two stacked spatial–spectral cross-attention encoders to extract interactive spatial–spectral fusion features. The feature embedding size for each encoder is set to 512.

GSC-ViT: This architecture consists of a spectral calibration block, two stacked feature extraction blocks, and a SoftMax classifier. Each feature extraction block comprises a GSC module and a groupwise separable ViT (GS-ViT) module.

SS-MTr: Unlike the above methods focusing on single-scale, the SS-MTr introduces a multiscale data augmentation module for data preprocessing. This framework consists of an encoder

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TABLE II
SENSITIVITY ANALYSIS FOR THE PROPOSED METHOD WITH DIFFERENT SIZES OF INPUT PATCHES ON THE INDIAN PINES, PAVIA UNIVERSITY, AND HOUSTON 2013 DATASETS

Patch Sizes	Indian Pines			Pavia University			Houston 2013		
	OA (%)	AA (%)	Kappa (%)	OA (%)	AA (%)	Kappa (%)	OA (%)	AA (%)	Kappa (%)
9×9	97.44	96.42	97.08	99.36	99.18	99.15	98.88	99.00	98.79
11×11	97.18	96.33	96.78	99.56	99.34	99.42	99.13	99.22	99.06
13×13	**98.7**	**98.46**	**98.52**	**99.75**	**99.57**	**99.67**	99.34	**99.39**	99.29
15×15	98.65	98.20	98.46	99.37	99.19	99.16	**99.35**	**99.39**	**99.30**
17×17	97.40	96.66	97.03	99.69	99.32	99.59	99.05	99.1	98.97

and a decoder, comprising four and three transformer blocks, respectively.

C. Parameter Analysis

In this section, a series of experiments are carried out to determine the optimal parameters for IGroupSS-Mamba, including the input patch sizes, the downsampling scales, the embedding dims, and the stacked depths.

1) Different Input Patch Sizes: Table II presents the impact of different input patch sizes on classification performance, ranging from 9×9 to 17×17 with a growth interval of 2. As can be observed, the Houston 2013 dataset displays a stable tendency of consistent increase followed by a decrease, and the Pavia University and Indian Pines datasets exhibit slight fluctuation, with their maximum peak at the identified point 13×13. Smaller patches may result in insufficient information, whereas larger patches can introduce redundant noise factors, leading to information aliasing. Consequently, a patch size of 13×13 is leveraged for all the datasets in our experiments.

2) Different Numbers of Principal Components in PCA: PCA transforms the original high-dimensional data into a set of orthogonal components, reducing the overall dimensionality by retaining the most informative variance. To determine the optimal number of principal components, experiments are conducted by truncating different numbers (L) of principal components. Table III depicts the classification performance and computational efficiency [model parameters and floating point operation (FLOPs)] on the Pavia University dataset, where the number of principal components is set to {10, 30, 50, 70, 90, all}. The *all* refers to retaining all the principal components without performing the PCA preprocessing step. As can be observed, incorporating PCA significantly outperforms the scenario without it, with quantitative improvements up to 10.6%, 11.92%, and 13.96% in terms of OA, AA, and Kappa, respectively. The performance initially exhibits progressive improvement as the number of principal components increases. However, the Hughes phenomenon occurs when the dimensionality exceeds 30, suffering from accuracy degradation. Therefore, we select the top 30 principal components for IGroupSS-Mamba, striking a balance between classification performance and computational efficiency.

3) Different Embedding Dims and Depths: To investigate the optimal structure of IGroupSS-Mamba for classification, a series of experiments are conducted by adjusting the feature embedding dimensions and the stage depths. Figs. 6 and 7 demonstrate the corresponding variations on three datasets in terms of OA (%), AA (%), and Kappa (%). As can be observed, the performance exhibits a progressively increasing trend with the growth of embedding dimensions and stage depths. However, excessive high dimensions and deeper depths provide limited accuracy improvements but computational complexity and burdens. By trading off these metrics, the proposed IGroupSS-Mamba is constructed as a lightweight structure, with the embedding dimension set to 32 and the stage depth determined as 3.

TABLE III
SENSITIVITY ANALYSIS WITH DIFFERENT NUMBERS OF PCA COMPONENTS ON THE PAVIA UNIVERSITY DATASET

PCA	Pavia University				
	OA (%)	AA (%)	Kappa (%)	Params (M)	Flops (G)
10	99.59	99.37	99.46	**0.0368**	**0.0046**
30	**99.75**	**99.46**	**99.66**	0.0573	0.0095
50	99.69	99.36	99.59	0.0777	0.0143
70	99.63	99.27	99.51	0.0982	0.0192
90	99.57	99.16	99.43	0.1187	0.0241
All	89.15	87.54	85.70	0.1320	0.0273

4) Different Downsample Scales: The downsampling operation is achieved by performing pixel aggregation along the spatial dimension. Fig. 8 investigates the impact of different aggregation scales on three datasets. Taking $[m, s] = [2, 1]$ as an example, the aggregation kernel is set to 2×2 and the aggregation stride is 1. From the results, different aggregation scales contribute variably to the extraction of multiscale spatial–spectral information, exhibiting slight performance fluctuations. In our experiments, the downsample scale [2, 1] is uniformly applied across all three datasets.

D. Experimental Comparison With Competitive Approaches

Tables IV–VI summarize the quantitative accuracies (OA (%), AA (%), and Kappa (%)) on the Indian Pines, Pavia University, and Houston 2013 datasets, with the optimal results highlighted in bold. The corresponding visualization maps are presented in Figs. 9–11.

1) Indian Pines Dataset: The experiments on the Indian Pines dataset are performed with 10% of the reference samples. The quantitative classification accuracies are reported in Table IV. As can be observed, the proposed IGroupSS-Mamba demonstrates significant superiority and effectiveness compared with other studied methods. Conventional SVM model and CNN-based methods including 2D-CNN and 3D-CNN are inherently constrained by manually crafted descriptors and local receptive fields, inevitably suffering from limited

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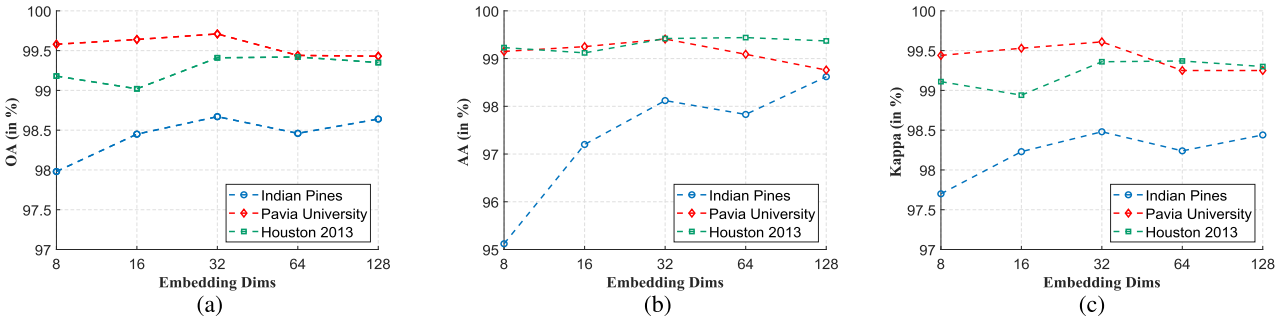


Fig. 6. Sensitivity analysis for the proposed method with different feature embedding dimensions in terms of (a) OA, (b) AA, and (c) Kappa.

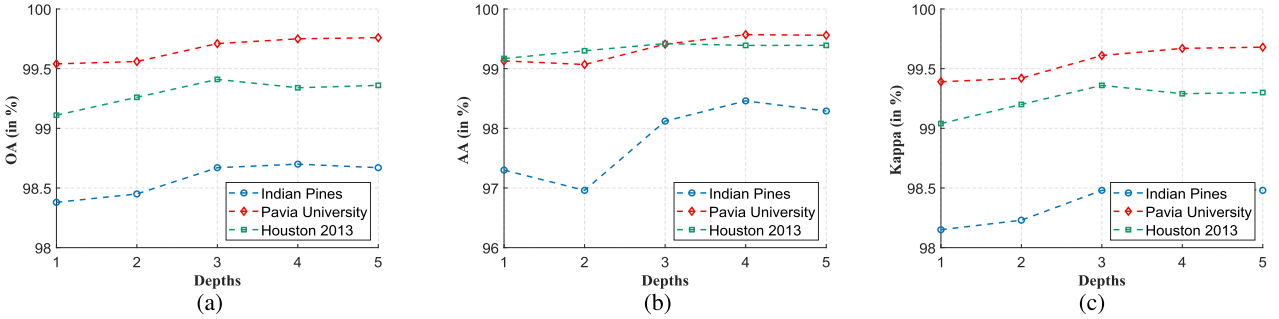


Fig. 7. Sensitivity analysis for the proposed method with different stage depths in terms of (a) OA, (b) AA, and (c) Kappa.

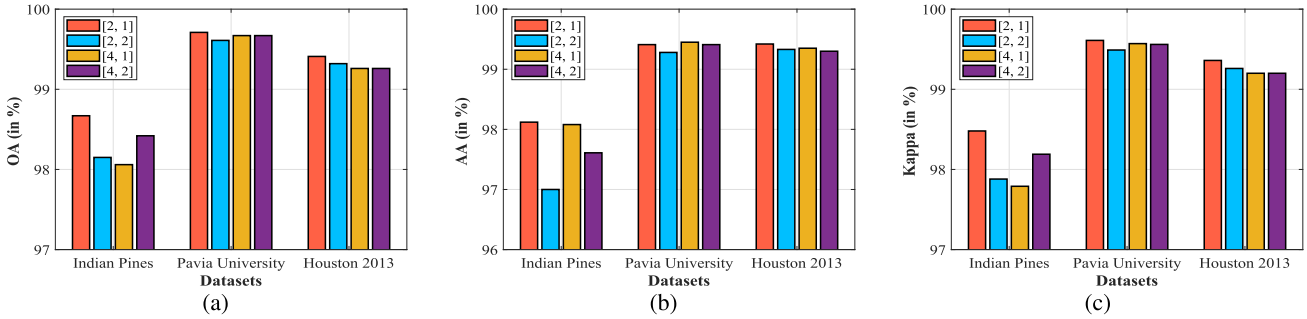


Fig. 8. Sensitivity analysis for the proposed method with different downsample scales in terms of (a) OA, (b) AA, and (c) Kappa.

feature extraction capabilities in handling complex data. Transformer-based architectures establish long-range dependencies based on the attention mechanism. The classical SF method focuses on the GSE to learn localized spectral representations, while ignoring the exploration of spatial contextual information. Compared with the proposed IGroupSS-Mamba, it demonstrates a substantial accuracy reduction of 8.04% in OA, 11.33% in AA, and 9.17% in Kappa. Recent SS-MTr and GSC-ViT methods integrate the grouping strategy and multiscale characteristics into the transformer network, respectively, which achieve slight performance advantages but remain inferior to IGroupSS-Mamba. Comparatively, IGroupSS-Mamba performs global spatial–spectral contextual modeling based on the selective state space models. The intervalwise feature grouping strategy allows the model to leverage the complementarity of different scanning directions, and the cascading of spatial and spectral operators further exploits the correlation between spatial and spectral contextual, which significantly achieves the comprehensive extraction of multiscale spatial–spectral semantic representations. Compared with the suboptimal GSC-ViT approach, the quantitative improvements in terms of OA, AA, and Kappa up to 0.43%, 3.91%, and 0.49%, respectively, were observed.

To further illustrate the differences in classification results, Fig. 9 presents the visualization maps of various comparison methods. It can be observed that SVM, 2D-CNN, 3D-CNN, and HSI-BERT generally suffer from significant noise interference and pronounced confusion in edge information. In contrast, the proposed IGroupSS-Mamba demonstrates the most consistent results with the ground truth, achieving clear category boundaries with minimal artifacts.

2) Pavia University Dataset: The classification experiments on the Pavia University dataset are conducted with 5% of the reference samples. Table V presents the quantitative comparison against various competitive approaches. As evidenced by the results, the proposed IGroupSS-Mamba achieves superior

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TABLE IV

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, κ, AND THE ACCURACIES OF EACH CLASS FOR THE INDIAN PINES DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Class	Conventional	CNN-based Methods		Transformer-based Methods						IGroupSS
	SVM	2D-CNN	3D-CNN	HSI-BERT	SF	SSTN	CASST	GSC-ViT	SS-MTr	-Mamba
1	20.73	76.09	95.65	48.78	100.00	76.07	78.26	53.66	**100.0**	**100.0**
2	73.07	92.16	88.31	78.44	78.37	93.63	97.34	**98.37**	94.19	95.49
3	62.99	87.23	84.94	80.58	89.29	97.35	97.71	99.33	98.80	**100.0**
4	50.70	64.98	87.76	51.17	81.22	78.06	99.58	97.18	92.41	**98.12**
5	92.64	96.48	96.48	91.24	89.89	99.17	92.96	**99.31**	98.34	99.08
6	94.90	98.77	96.16	96.19	97.41	98.63	97.81	**100.0**	**100.0**	99.24
7	76.00	32.14	78.57	20.00	84.00	96.43	96.43	**100.0**	**100.0**	**100.0**
8	96.63	**100.0**	**100.0**	98.83	**100.0**	**100.0**	99.79	**100.0**	**100.0**	**100.0**
9	33.33	30.00	70.00	0.00	27.78	65.00	65.00	83.33	85.00	**94.44**
10	68.69	91.87	92.80	78.37	94.97	97.84	92.80	98.97	**99.59**	98.29
11	85.16	95.93	91.20	92.85	93.44	98.04	98.37	98.37	**99.59**	99.55
12	64.89	93.42	88.03	62.47	81.65	88.53	91.40	96.25	94.10	**97.38**
13	97.03	97.56	94.15	96.19	87.03	98.54	91.71	**100.0**	98.54	98.92
14	96.49	98.81	99.05	94.46	94.82	99.84	99.76	99.91	99.92	**100.0**
15	55.33	81.35	89.38	87.89	96.83	99.48	95.34	90.78	93.01	**100.0**
16	93.93	97.85	**100.0**	65.06	95.24	88.17	81.72	95.24	88.17	92.86
OA (%)	79.82	93.34	92.17	85.45	90.67	96.56	96.65	98.28	97.92	**98.71**
AA (%)	72.03	83.41	90.78	71.41	87.00	92.18	92.25	94.42	96.35	**98.33**
Kappa	76.84	92.39	91.08	83.36	89.36	96.07	96.18	98.04	97.63	**98.53**

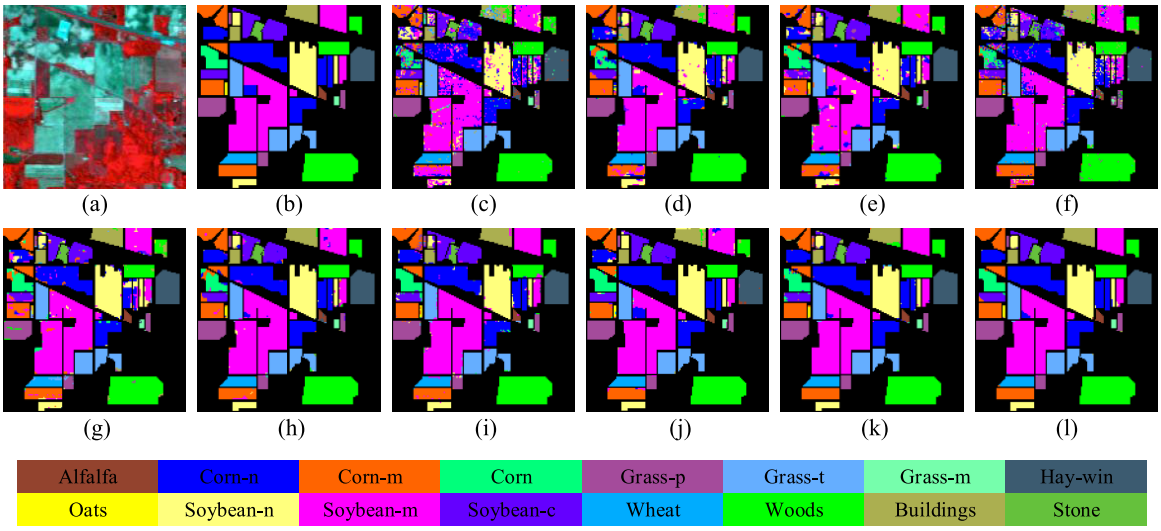


Fig. 9. Classification maps obtained by the compared methods on the Indian Pines dataset. (a) Original data. (b) Reference map. (c) SVM. (d) 2D-CNN. (e) 3D-CNN. (f) HSI-BERT. (g) SF. (h) SSTN. (i) CASST. (j) GSC-ViT. (k) SS-MTr. (l) IGroupSS-Mamba.

TABLE V

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, κ, AND THE ACCURACIES OF EACH CLASS FOR THE PAVIA UNIVERSITY DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Class	Conventional	CNN-based Methods		Transformer-based Methods						IGroupSS
	SVM	2D-CNN	3D-CNN	HSI-BERT	SF	SSTN	CASST	GSC-ViT	SS-MTr	-Mamba
1	91.63	96.31	95.94	97.63	94.75	99.52	99.17	99.63	99.32	**99.86**
2	97.56	99.37	99.71	99.93	98.14	**99.98**	99.95	99.90	99.90	99.94
3	73.90	82.22	89.76	86.51	84.70	92.04	98.28	97.32	97.57	**98.40**
4	92.10	94.35	97.36	97.53	96.81	**99.35**	93.08	98.29	96.54	99.14
5	98.51	99.85	**100.0**	99.61	99.22	**100.0**	98.66	99.89	99.78	99.61
6	86.04	93.65	97.41	93.05	94.37	99.24	99.60	99.74	**100.0**	**100.0**
7	84.09	89.17	96.69	95.80	84.56	99.85	99.62	99.59	99.32	**100.0**
8	90.25	88.73	93.92	97.86	92.65	94.95	94.73	98.63	92.12	**99.77**
9	99.22	97.57	**100.0**	99.67	**100.0**	99.79	96.41	98.93	93.35	98.44
OA (%)	92.75	95.77	97.62	97.61	96.60	98.95	98.63	99.46	98.63	**99.75**
AA (%)	90.37	93.47	96.75	96.40	93.91	98.30	97.72	99.10	97.55	**99.46**
Kappa	90.35	94.37	96.84	96.82	94.17	98.60	98.19	99.28	98.19	**99.66**

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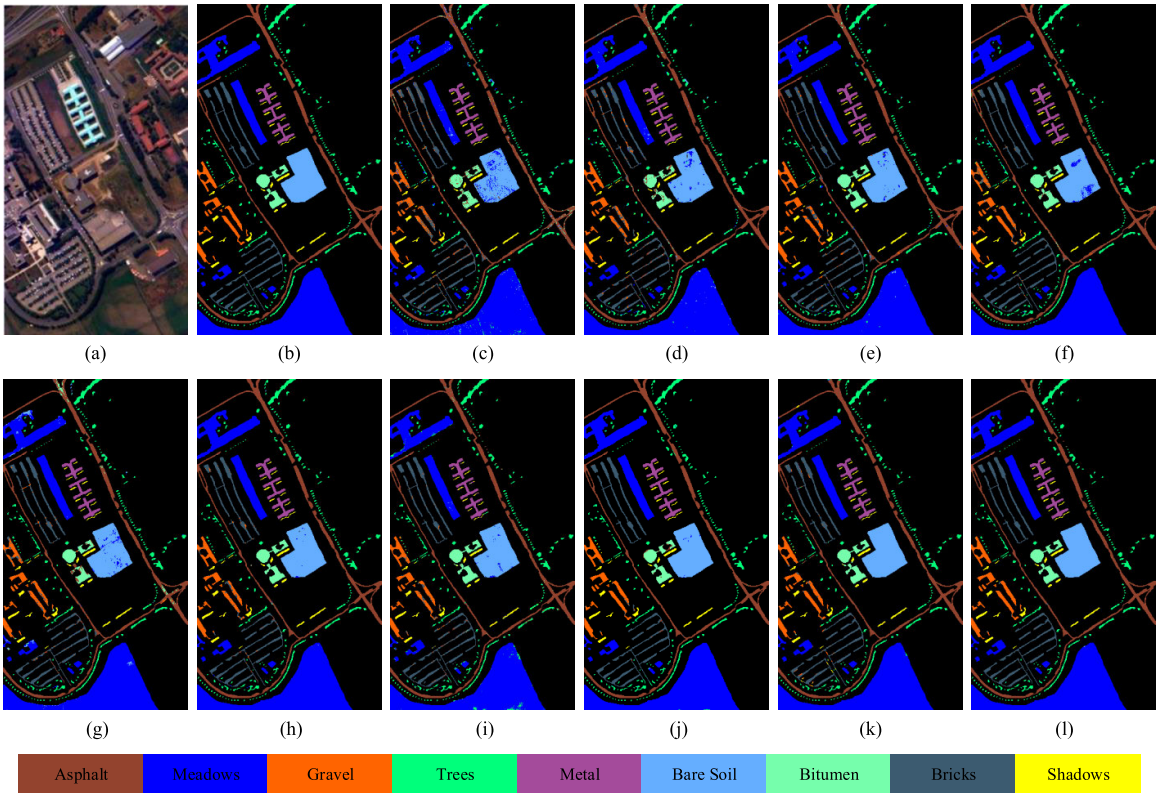


Fig. 10. Classification maps obtained by the compared methods on the Pavia University dataset. (a) Original data. (b) Reference map. (c) SVM. (d) 2D-CNN. (e) 3D-CNN. (f) HSI-BERT. (g) SF. (h) SSTN. (i) CASST. (j) GSC-ViT. (k) SS-MTr. (l) IGroupSS-Mamba.

TABLE VI

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, κ, AND THE ACCURACIES OF EACH CLASS FOR THE HOUSTON 2013 DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Class	Conventional	CNN-based Methods		Transformer-based Methods						IGroupSS
	SVM	2D-CNN	3D-CNN	HIS-BERT	SF	SSTN	CASST	GSC-ViT	SS-MTr	-Mamba
1	98.53	98.40	97.04	92.26	99.29	99.76	97.04	98.67	98.80	**99.91**
2	98.05	99.20	98.48	83.15	98.05	99.68	98.01	99.73	99.60	**99.82**
3	98.88	**100.0**	**100.0**	99.20	**100.0**	**100.0**	98.71	99.68	**100.0**	99.84
4	97.81	98.31	98.95	93.47	96.43	**99.84**	99.04	98.84	99.44	98.48
5	98.57	99.28	99.44	99.73	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
6	92.64	92.62	**100.0**	78.76	95.89	96.00	95.38	**100.0**	98.15	**100.0**
7	91.98	94.01	93.38	91.06	96.06	96.92	95.90	95.79	**98.10**	97.46
8	91.96	89.63	94.29	86.68	91.70	97.67	96.22	97.59	96.70	**99.38**
9	85.09	83.87	93.61	89.60	94.59	97.36	95.53	91.66	98.80	**99.73**
10	93.16	94.62	96.33	89.03	97.46	99.51	99.27	99.18	99.84	**100.0**
11	87.01	90.53	96.84	95.31	90.56	99.67	97.89	99.01	98.79	**100.0**
12	87.25	89.38	95.46	88.63	99.28	**99.68**	98.54	98.02	97.89	99.55
13	32.35	77.83	94.24	93.12	56.40	82.83	97.23	97.39	88.81	**97.63**
14	98.91	99.30	98.60	97.92	98.44	99.53	**100.0**	99.74	**100.0**	**100.0**
15	85.08	**100.0**	99.85	99.66	99.66	**100.0**	99.39	**100.0**	**100.0**	**100.0**
OA (%)	91.74	93.93	96.77	91.67	95.46	98.54	97.85	98.11	98.62	**99.45**
AA (%)	90.03	93.80	97.10	91.84	94.25	97.89	97.88	98.35	98.34	**99.45**
Kappa	91.06	93.44	96.51	90.99	95.09	98.42	97.68	97.95	98.50	**99.40**

classification performance, particularly outperforming other methods in several categories such as Asphalt, Bitumen, and Self-blocking bricks. In contrast to the GSC-ViT method, IGroupSS-Mamba demonstrates notable improvements in OA, AA, and Kappa by 0.29%, 0.36%, and 0.38%, respectively, further revealing the effectiveness of the grouping-based multidirectional scanning mechanism and multiscale extraction strategy. In addition, the corresponding classification maps generated by comparison approaches are visualized in Fig. 10. Due to the constrained feature extraction capability of handcrafted descriptors, the conventional SVM method exhibits notable misclassification, particularly evident in the bare soil category. In contrast, the proposed IGroupSS-Mamba delivers the clearest and smoothest classifications across most regions.

3) Houston 2013 Dataset: The experiments on the Houston 2013 dataset are executed with 10% of the labeled samples.

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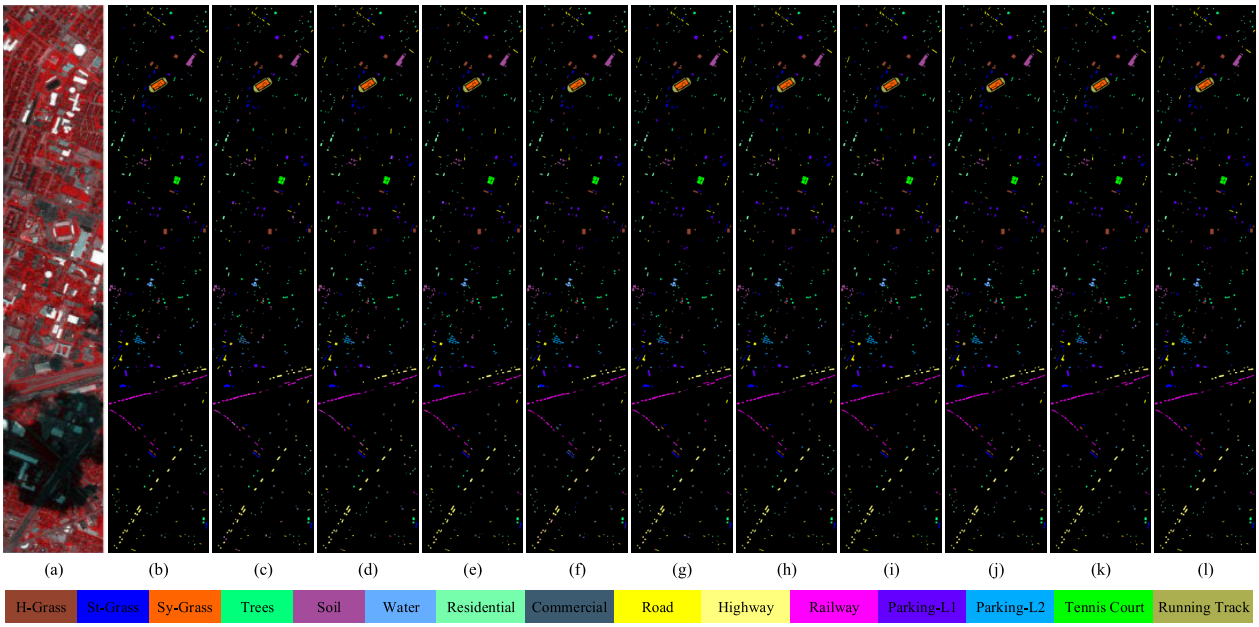


Fig. 11. Classification maps obtained by the compared methods on the Houston 2013 dataset. (a) Original data. (b) Reference map. (c) SVM. (d) 2D-CNN. (e) 3D-CNN. (f) HSI-BERT. (g) SF. (h) SSTN. (i) CASST. (j) GSC-ViT. (k) SS-MTr. (l) IGroupSS-Mamba.

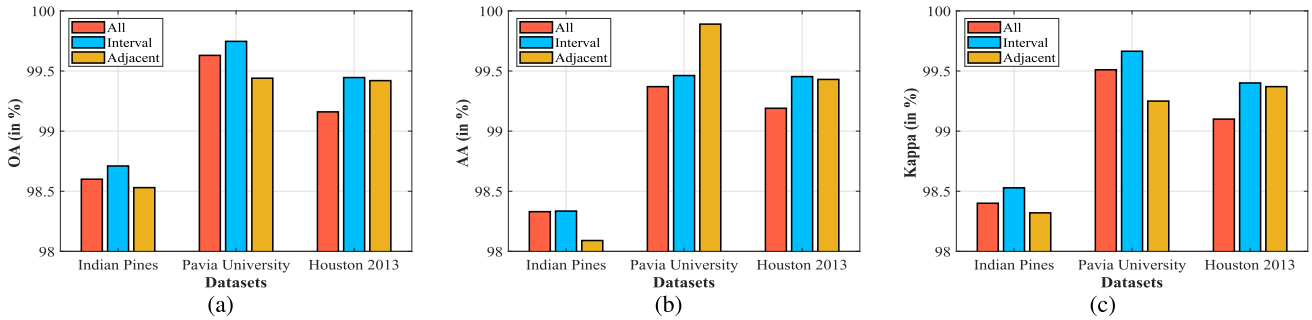


Fig. 12. Effectiveness analysis of different grouping strategies on three datasets in terms of (a) OA, (b) AA, and (c) Kappa.

The classification results are illustrated in Table VI. As evident from the results, the proposed IGroupSS-Mamba consistently outperforms other competitive approaches by substantial margins, demonstrating the highest quantities across all three metrics. Restricted by the modeling capability, the suboptimal SS-MTr inevitably exhibits degraded classification consequences, with the reduction compared with IGroupSS-Mamba reaching 0.83% for OA, 1.11% for AA, and 0.9% for Kappa, respectively. According to the visualization maps in Fig. 11, IGroupSS-Mamba provides the most precise prediction details, despite the predominantly discrete and localized sample targets in this scenario. These significant improvements further validate the potential of SSMs in HSI classification.

E. Ablation Study

1) Effectiveness of Interval Grouping Strategy: Fig. 12 illustrates the performance variations with different grouping strategies on three HSI datasets. The All denotes applying multidirectional scanning into all spectral bands without grouping. The Adjacent denotes performing feature grouping in an adjacent manner, generating four local HSI cubes with continuous features. The Interval refers to the proposed interval grouping strategy as (10), constructing the nonoverlapping feature groups in an interval manner. As can be observed, the proposed Interval strategy demonstrates more competitive superiority and advantages in HSI classification, while the All and Adjacent schemes exhibit slight performance degradations and fluctuations. The potential reason might be that the All strategy encounters challenges with information redundancy in sequence modeling, which arises from the excessively high spectral dimensionality. The Adjacent scheme tends to capture local information of the spectrum, as depicted in Fig. 1(b), thereby constraining its ability to adequately explore global spatial–spectral correlations. Comparatively, the Interval strategy accounts for the similarity between adjacent spectral bands, allowing each feature group to comprehensively preserve global characteristics without redundancy, as illustrated in Fig. 1(c). These notable advantages significantly reveal the feasibility and superiority of the proposed interval grouping strategy.

2) Effectiveness of Spatial and Spectral Operators: To highlight the specific effectiveness of the spatial and spectral operators in IGSSB, ablation experiments are conducted on

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TABLE VII
EFFECTIVENESS ANALYSIS OF SPATIAL AND SPECTRAL OPERATORS ON THE INDIAN PINES, PAVIA UNIVERSITY, AND HOUSTON 2013 DATASETS

	Indian Pines			Pavia University			Houston 2013		
	OA (%)	AA (%)	Kappa (%)	OA (%)	AA (%)	Kappa (%)	OA (%)	AA (%)	Kappa (%)
Spatial Operator	98.56	97.69	98.35	99.66	99.29	99.55	99.41	99.43	99.36
Spectral Operator	98.42	97.84	98.20	99.65	99.25	99.53	99.35	99.38	99.30
Parallel Spa-Spe	98.43	97.44	98.21	99.55	99.18	99.40	99.29	99.32	99.23
Cascaded Spe-Spa	98.63	98.17	98.43	99.72	99.43	99.62	99.25	99.28	99.19
Cascaded Spa-Spe	**98.71**	**98.33**	**98.53**	**99.75**	**99.46**	**99.66**	**99.45**	**99.45**	**99.40**

TABLE VIII
EFFECTIVENESS ANALYSIS OF ATTENTION MECHANISM ON THE INDIAN PINES, PAVIA UNIVERSITY, AND HOUSTON 2013 DATASETS

	Indian Pines			Pavia University			Houston 2013		
	OA (%)	AA (%)	Kappa (%)	OA (%)	AA (%)	Kappa (%)	OA (%)	AA (%)	Kappa (%)
w/o Attention	98.54	97.87	98.33	99.40	99.04	99.21	99.06	99.16	98.98
IGoupSS-Mamba	**98.71**	**98.33**	**98.53**	**99.75**	**99.46**	**99.66**	**99.45**	**99.45**	**99.40**

three HSI datasets by removing either the spectral or spatial into a single-operator setup. The classification results in terms of accuracy metrics are illustrated in Table VII. It can be observed that the individual Spatial Operator and Spectral Operator each achieve significant classification performance, revealing the effectiveness of the proposed scanning directions along the spatial and spectral dimensions. In addition, the cascade of spatial and spectral operators further enhances the classification capability, demonstrating competitive advantages and strong robustness across all datasets. Specifically, the Cascaded Spa-Spe surpasses the Spatial Operator by an average of 0.09% in OA, 0.28% in AA, and 0.11% in Kappa. It exceeds the Spectral Operator by margins of 0.14%, 0.26%, and 0.14% for OA, AA, and Kappa, respectively. These disparities significantly accentuate the superiority of the cascading structure, which adequately takes advantage of the complementarity of spatial and spectral information of hyperspectral data.

In addition, this section investigates the effectiveness of various combination structures for the Spatial Operator and Spectral Operator, including Cascaded Spa-Spe, Cascaded Spe-Spa, and Parallel Spa-Spe. As observed in Table VII, all the structures achieve competitive classification performance, demonstrating the superiority of the constructed operators in modeling global spectral–spatial contextual relationships. Notably, the Cascaded Spa-Spe showcases the optimal performance across all three datasets, while the Cascaded Spe-Spa exhibits slight fluctuations. The Parallel Spa-Spe adopts the "divide and conquer" principle, using a parallel structure for these two operators along with an addition operation. Taking the Indian Pines dataset as an example, the Parallel Spa-Spe structure is inferior to the Cascaded Spa-Spe by margins of 0.28%, 0.89%, and 0.32% for OA, AA, and Kappa, respectively. The possible reason might be that the cascade operation facilitates more coherent information integration and stable model training compared with the parallel structure, leading to better convergence and higher classification accuracy.

3) Effectiveness of Attention Mechanism in IGSM: This section systematically evaluates the classification performance with and without the attention mechanism on three HSI datasets. The results in terms of OA, AA, and Kappa are illustrated in Table VIII. It is evident that the attention mechanism plays an important role in the feature interval concatenation procedure. Compared with the IGroupSS-Mamba without attention mechanism, the classification performance demonstrates notable improvements, with average increases of 0.3% in OA, 0.39% in AA, and 0.35% in Kappa. By modulating channelwise importance with dynamic weights, the attention mechanism significantly encourages intergroup information interaction during the aggregation process, enhancing the model's ability to capture global spatial–spectral semantics while reducing the influence of redundant information.

F. Robustness Assessment

To assess the robustness of the proposed IGroupSS-Mamba framework, extensive experiments are performed by varying the proportions of training samples. Fig. 13 illustrates the performance variations in the three HSI datasets, with the proposed IGroupSS-Mamba prominently highlighted by red curves. The selected percentage for the Indian Pines and Houston 2013 datasets range from 1.0% to 10.0% in increments of 1.0%, while for the Pavia University dataset, the percentages are set at 0.5%, 1.0%, up to 5.0% in increments of 0.5%. From the results, the classification accuracy of IGroupSS-Mamba achieves a steady enhancement trend as the proportion of training samples increases. Benefiting from the interval grouping strategy and S6 sequence modeling mechanism, IGroupSS-Mamba significantly outperforms other competitive methods across most percentages on the three testing datasets, which further demonstrates the robustness and superiority of the proposed method.

G. Feature Visualization

In addition, t-SNE [47] is used to visualize the learned feature distributions of the proposed IGroupSS-Mamba, providing an intuitive interpretability of the classification process. As depicted in Fig. 14, a number of target samples from each dataset are randomly selected for visualization. As can be observed, IGroupSS-Mamba exhibits effective separation between different categories, while maintaining significant compactness within the same class. Leveraging the interaction between the interval grouping strategy and SSMs,

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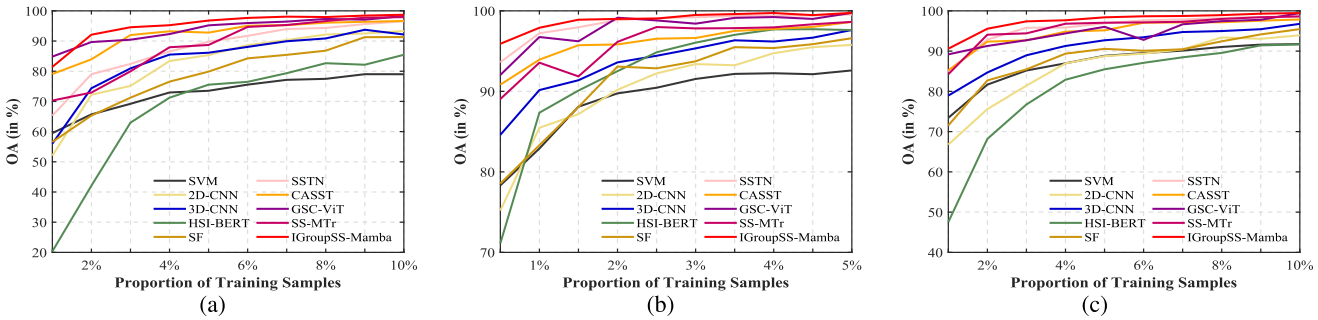


Fig. 13. Classification performance of the compared methods with different proportions of labeled data on three datasets (a) Indian Pines. (b) Pavia University. (c) Houston 2013.

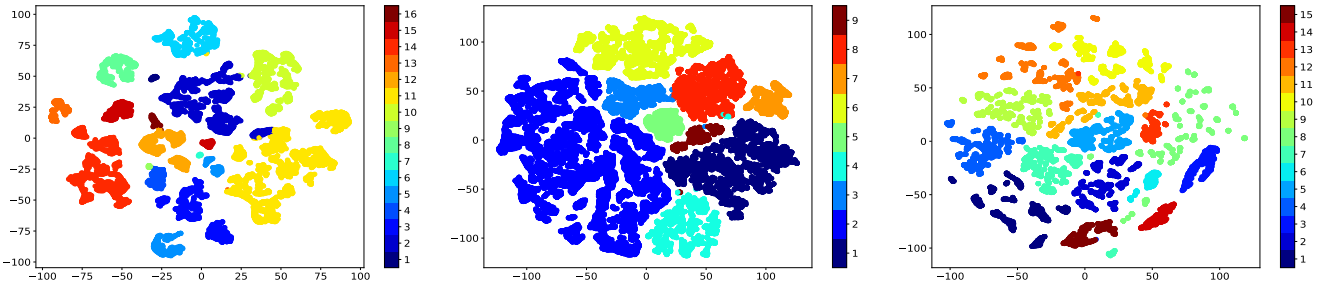


Fig. 14. Visualization of the learned features of the proposed IGroupSS-Mamba on (a) Indian Pines. (b) Pavia University. (c) Houston 2013.

TABLE IX
COMPUTATIONAL PARAMETER ANALYSIS OF VARIOUS COMPARISON APPROACHES IN TERMS OF PARAMETERS, FLOPS, AND INFERENCE TIME ON THE PU DATASET

	Metrics	2D-CNN	3D-CNN	HIS-BERT	SF	SSTN	CASST	GSC-ViT	SS-MTr	IGroupSS -Mamba
Indian Pines	Params (M)	0.0173	0.0165	0.3951	0.1226	0.3615	6.5275	0.5636	1.6153	0.0575
	Flops (G)	0.0003	0.0024	7.1523	0.0245	0.0033	0.9893	0.0201	0.1199	0.0095
	Inference time (s)	0.193	0.27	9.021	2.139	0.554	3.23	1.64	1.36	4.79
Pavia University	Params (M)	0.0101	0.0073	0.3006	0.1222	0.3577	6.5275	0.0778	1.4622	0.0573
	Flops (G)	0.0001	0.0012	5.441	0.0126	0.0032	0.9893	0.0049	0.1197	0.0095
	Inference time (s)	0.56	0.702	36.13	6.55	2.18	13.4	13.256	5.29	19.76
Houston 2013	Params (M)	0.0132	0.0123	0.3773	0.1133	0.3609	6.5316	0.0887	1.3667	0.0575
	Flops (G)	0.0002	0.0017	6.8135	0.0163	0.0032	0.9893	0.0056	0.0377	0.0095
	Inference time (s)	0.247	0.313	12.218	1.773	0.78	161.34	45.41	1.11	3.30

IGroupSS-Mamba achieves multidirectional and multiscale global spatial–spectral information extraction. This characteristic facilitates the model to learn the most discriminative features, thereby constructing the robust class boundaries for the testing data.

H. Analysis of Computational Complexity

As illustrated in Table IX, the computational complexity of the proposed IGroupSS-Mamba on all the datasets is detailed, focusing on model parameters, Flops, and inference time. Parameters represent the number of parameters in the entire network. Flops signifies the number of floating-point operations performed by the model. Although CNN-based methods exhibit relatively low computational burdens, their ability to capture long-range dependencies is limited by local receptive fields. Transformer-based architectures generally suffer from higher resource consumption due to the inherent MHSA modules. Benefiting from the linear sequential modeling mechanism of SSMs and feature grouping strategy, the proposed IGroupSS-Mamba consistently demonstrates reduced computational parameters and Flops across various datasets. The inference time is also comparatively acceptable compared with other approaches. In summary, IGroupSS-Mamba presents a competitive balance between computational efficiency and classification performance, underscoring significant potentiality and viability for HSI classification.

V. CONCLUSION

In this article, we introduce an IGroupSS-Mamba framework, which supports multidirectional and multiscale spatial–spectral information processing in a grouping and hierarchical manner. Benefiting from the IGSM, IGroupSS-Mamba achieves nonredundant spatial–spectral dependencies modeling across different feature groups, which enjoys the

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complementarity of different scanning directions while alleviating information interference. Meanwhile, the design of IGSSB enables IGroupSS-Mamba to effectively leverage the complementary strengths of spectral and spatial information inherent in hyperspectral data. Extensive experiments indicate that IGroupSS-Mamba efficiently overcomes the performance and efficiency limitations in state-of-the-art CNN-based and transformer-based HSI architectures, which provides a feasible solution for HSI classification task. Future research will focus on integrating the Mamba model with alternative network architectures and extending its application to more challenging hyperspectral scenarios.

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